Srinivasan Poonkundran

(520) 365-5332 | Tucson, AZ (Open to Relocation) | srinivasan.poonkundran@gmail.com | LinkedIn | GitHub | Portfolio

Professional Experience

Research Assistant — Feb. 2026 – Present
The University of Arizona — *Tucson, AZ*

- Built and evaluated machine learning pipelines for large-scale sensor and behavioral datasets using Python and scikit-learn.
- Implemented anomaly detection workflows using Isolation Forest, Local Outlier Factor (LOF), and deep autoencoder-based models for time-series analysis.
- Developed reproducible preprocessing, evaluation, and experimentation pipelines for data-driven research workflows.
- Worked with large-scale time-series datasets containing over 1M+ records for behavioral analysis and anomaly detection studies.

Programmer Analyst — Full-Stack Software Engineer — Sept. 2021 – Jan. 2024
Cognizant Technology Solutions — *Remote*

- Built Java Spring Boot microservices deployed on AWS, backed by MySQL databases, secured with OAuth2-based authentication and authorization, and integrated with Angular dashboards used by internal stakeholders.
- Improved API response time by 35% by optimizing SQL queries and caching critical endpoints.
- Built a Python-based PDF label validation tool to automate manual checks, reducing operational effort and saving approximately $1.6K annually.
- Contributed to code reviews and refactored backend services with Java and Spring Boot, improving code quality and reducing defect occurrences, which accelerated release cycles.

Software Developer — Jan. 2023 – Jan. 2024
Self-Employed — *Part-time, Remote*

- Designed and deployed a production Android application supporting 1,000+ monthly transactions for a retail client.
- Built real-time billing, sales tracking, and revenue analytics dashboards within the mobile interface.
- Integrated AWS S3 for secure invoice storage, backup, and cross-device data retrieval.
- Implemented robust local storage and synchronization mechanisms to ensure data reliability under unstable network conditions.

Projects

GradeLens – RAG-Powered AI Agent for Intelligent Evaluation | GitHub

- Architected a production-grade Retrieval-Augmented Generation (RAG) system using vector embeddings and LLM orchestration to enable scalable, context-aware automated grading.
- Designed modular ML pipelines for embedding generation, semantic retrieval, rubric validation, and structured output enforcement.
- Implemented asynchronous API handling with parallel execution to reduce grading latency across multiple exam questions.
- Deployed the system within a Django + React application, enabling scalable, low-latency inference.

Emotion Detection from Text | GitHub

- Designed and trained a transformer-based NLP classification system using BERT embeddings and Bi-GRU layers, achieving 0.83 F1-micro across 7 emotion classes.
- Built scalable preprocessing and training pipelines using Hugging Face and TensorFlow to efficiently handle large text datasets.
- Optimized model performance through structured tuning and regularization to improve generalization.

Education

The University of Arizona — Jan. 2024 – Dec. 2025
Master of Science, Data Science — *Tucson, AZ*
Honors: Distinguished Graduate Scholar (Fall 2025)

Savitribai Phule Pune University — Aug. 2017 – May 2021
Bachelor of Engineering, Computer Engineering — *Pune, India*

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, R, C, C++
Development Practices: Object-Oriented Programming, Data Structures & Algorithms, Microservice Architecture, Agile Methodologies, SDLC, CI/CD, Maven, Unit Testing (JUnit, Mockito)
Database & Cloud: MySQL, MongoDB, SQL/NoSQL, AWS - EC2, ECS, Load Balancers, Elastic Beanstalk, S3, IAM
AI Systems: Agno, LangChain, Retrieval-Augmented Generation (RAG), Vector Databases (pgvector)
Certifications: Amazon Web Services Certified Cloud Practitioner | Credentials